Ulam and Rota discuss von Neumann

The following discussion between Stanisław Ulam and Gian-Carlo Rota about Johnny von Neumann took place in Gainesville, Florida in 1974.

We have made a few small editorial changes to the text.

ROTA : Von Neumann was older than you.

ULAM : Six, seven years.

ROTA : An older man!

ULAM : Yes. You know how it is. In the beginning the percentage was twenty or so; later it went down to ten.

ROTA : So you considered him a senior, and yet you made fun of him?

ULAM : Oh always! I was always impudent.

ROTA : He did not treat you as someone younger?

ULAM : No. I don't think he knew anybody more intimately and vice versa, despite our difference in age. For a man of his stature he was curiously insecure, but his understanding, intelligence, mathematical breadth, and appreciation of what mathematics is for, historically and in the future, was unsurpassed. His immense work stands at the crossroads of the development of exact sciences. The rationalisation of the idea of infinity - the life blood of its history - with its mysterious power to encode succinctly and generally the properties of numbers and the patterns of geometry, received some of its definite formulations from his work. His ideas also advanced immeasurably the attempts to formalise the new, strange world of physics in the philosophically strange work of quantum theory. Fundamental ideas of how to start and proceed with the formal modes of operations and the scope of computing machines owe an immense debt to his work, though they still today give hints that are only dimly perceived about the workings of the nervous system and of the human brain itself.

Other mathematicians strike me as virtuosi who play their own special instruments. None are comparable to Johnny.

By the way, you were supposed to ask about the foods von Neumann liked.

ROTA : List the foods von Neumann liked and those he did not like!

ULAM : He was not a gourmet, but he liked to eat. He liked to go to restaurants, mainly, I think, to escape from the usual scene or routine. It was an excuse for not working, because he was a very hard worker. At home he worked at a desk, writing, a thing which irritated me a bit. When I stayed at his house and saw him suddenly leave to go upstairs and write, I, cruelly and foolishly I must say, would make fun of it. So for relaxation he liked to drive out for dinner. In Princeton we often went to a restaurant called Marot, on the highway to Trenton.

He never smoked, but he ate voluminously, which accounted for his increasing rotundity and portliness as the years went by. Sometimes when Klari, his second wife, could not finish what was on her plate, she would give it to Johnny or to me and say, "Both of you are human garbage cans!" Klari, by the way, was a very intelligent, very nervous woman who had a deep complex that people paid attention to her only because she was the wife of the great von Neumann, which was not true of course.

Johnny liked Mexican food, hot peppery stuff. I suspect it was because if he had a stomach ache later, he would know what to blame it on! I always have such Machiavellian suspicions. It is probably just that he was used to Hungarian goulashes and hot paprika. He liked sweets too, but on the whole what he wanted was volume, like me, like you too. You like the volume of pasta.

He had this nervous trait, an almost automatic response. For example, whenever he saw the words chicken mole on a menu, he would automatically intone Moles Hadriani, and I would respond Jacques de Molay - you know, the Grand Master of the Knights Templar. It was a game of association, just like you always add Pal [Hungarian for Paul] when you hear the word Erdős!

He also had occasionally an infrequent but noticeable stutter. He would say a word and repeat it two or three times in quick succession. I wonder whether it could have been an incipient physical lesion, for he died of things affecting his brain. Actually, on second thought it could not, because his cancer started somewhere else. Sometimes I suspect that his stutter was in order to gain time while thinking over a riposte or considering quickly some other angle for a statement, like a person lighting a pipe to gain time.

ROTA : How long did you know von Neumann?

ULAM : I first met him in Warsaw in 1935, but I had already started corresponding with him the year before, and that is when he invited me to visit him at the Institute in Princeton.

ROTA : What was he working on at Los Alamos during the war?

ULAM : On everything. He was one of the originators, one of the "influencers" of implosion. By the way, you are my most eminent "influencee"; it is a relationship different from teacher-student.

He worked on the whole project, scientifically and politically, especially with the hydrogen work.

ROTA : But actual work?

ULAM : Of course, mostly hydrodynamics.

ROTA : Did he know much physics?

ULAM : To some extent, but he did not have the physicist's feeling for experiment. His interest was more modern than Hilbert's. His interest was in the foundations of quantum mechanics, which were mathematical. And that could be taken as an example of mathematics not really useful for real physics.

But there was no bullshit in him. That is an expression he used about certain people. He would say, "It is very rare, but there is no bullshit in so-and-so."

Of course he worked, in answer to your question. In fact he was unable to play the role of senior scientist or advisor without being actively engaged, like with computing. Even towards the end of his life, when he was chairman of the Intercontinental Ballistic Missile (ICBM) Committee, a committee established by the President after Sputnik.

ROTA : I still don't have a picture of von Neumann's personality.

ULAM : He loved jokes, though I don't think he invented many, but he remembered and repeated them, and occasionally he made original and very witty remarks or saw comparisons which were comical. Most are unprintable.

A propos of the church knowing about the atom bomb, he said, "Priests will bless the active cores." And when he noticed all the churches of Los Alamos, he was much amused when I pointed to one church and called it "San Giovanni delle Bombe"! One of the first solid non-wood buildings in Los Alamos was built for the offices of the Atomic Energy Commission (AEC). He called it "El Palacio de la Seguridad"!

Oh! One thing about Johnny, he tended to tell people what they wanted to hear. He also used to tell me his little tactical discoveries. Once he said, "In Los Alamos it is very difficult to introduce novelty, but once introduced, it is impossible to get rid of it!"

After the war he was for a Pax Americana, and one could probably have established it, but the historical perspective, the desire to do it were not present in the country. The general population was not thinking in those terms. Although, when World War II ended, Americans were like Roman citizens during the Roman Empire. By commuting through the American bases one could go anywhere in Europe without encountering the native populations. This was really a beginning of that sort of thing, but for good or for bad - who knows - it quickly dissipated.

What else would you like to know about von Neumann?

ROTA : Always well dressed, wasn't he?

ULAM : Not really well dressed, but simple, decent, well-cut, classic city dress.

ROTA : I still don't have a picture of the man.

ULAM : He became an important government figure and very influential in ballistic missile development.

ROTA : It is strange how you like every thing about him except his work in mathematics.

ULAM : Really? No, not quite so. But he was not a mathematician's mathematician. He did little in number theory, some in continuous geometry and operators and Hilbert space, and some in measure theory and group theory.

To my mind and to my taste, the most important work he did is what he did when he was getting older, which mathematicians don't appreciate, namely his speculations on automata, on the brain, and his contributions to computing and to problems in hydrodynamics.

He knew about quantum theory and some parts of theoretical physics, which few mathematicians did. He contributed to the grammar of physics, so to say. One must also mention the theory of games. What interested me less was his work in the almost-periodic functions of groups.

ROTA : Can you tell me something about how his mind worked?

ULAM : It is curious to me that in our many mathematical conversations on topics belonging to set theory and allied fields, he always seemed to think formally. Most mathematicians, when discussing problems in these fields, seem to have an intuitive framework based on geometrical or almost tactile pictures of abstract sets, transformations, and such. Johnny gave the impression of operating sequentially by formal deductions. His intuitions seemed very abstract; they involved a complementarily between the formal appearance of a collection of symbols, the games played with them, and the interpretation of their meanings. Something like the distinction between a mental picture of the physical chess board and a mental picture of a sequence of moves on it written down in algebraic notation!

The quickness of his thinking was quite remarkable. He saw immediately the possibilities of Monte Carlo. To my mind this was much more important than one hundred papers in partial differential equations! It is at least a general procedure - I would not quite call it a method - and he invented many tricks for it and specific ways to get random distributions. It was very pleasant to discuss it with him.

Too bad he did not live to see how computers have revolutionised everything and what influence they will have on science in general and even on pure mathematics. His role in their development was tremendous, and if I may say so I would say I too played a modest role in showing how to use computers!

ROTA : How would you characterise his influence?

ULAM : There used to be a time when there were mathematicians who gave specific ideas and choice of topics and directions either explicitly or by implication to the work of other mathematicians. Not to go back centuries but less than a hundred years, let us say Poincaré, Hilbert, in more recent

times Herman Weyl. Hilbert had laid what was hoped would be a foundation for the final axiomatisation of mathematics and beyond, of all science. Little did he know that in the thirties the unavoidable limitations of this approach would be revealed.

Von Neumann was one of these giants too in the breadth of his knowledge, especially when one remembers that now the diversity and complexity of contemporary problems enormously surpass the situation confronting Poincaré and Hilbert. Yet, he admitted to me that he felt he did not know even a third of mathematics, that he did not think it was possible nowadays for any one brain to have more than a passing knowledge of more than one-third of pure mathematics.

So, at his suggestion and for his amusement I concocted an oral doctoral examination in various fields in such a way that he would not be able to pass it. And indeed, when I thought about what problems to give him in each domain, I found one in differential geometry, one in number theory, one in algebra and a couple of others. And he agreed that he could not have answered any of the questions and the exam would have been a complete failure. Which goes to show that doctoral exams are to some extent meaningless. Of course, if one prepares for some specific topics, that is something else.

ROTA : Who was von Neumann a student of?

ULAM : He considered himself a student of Erhardt Schmidt. It was not easy for me to get to the bottom of this. One reason, I suspect, is that Schmidt did some work in combinatorics which always interested Johnny very much.

ROTA : It was the Hilbert space. Schmidt was the only person at the time who studied nonlinear operators.

ULAM : But Johnny did not.

ROTA : That is why he admired Schmidt!

ULAM : Also I remember that he told me that Schmidt did not like to write. That surprised Johnny. I also think he secretly admired it. He said that Schmidt had told him that he felt faint whenever he saw a blank sheet of paper. Johnny was not at all like that. On the contrary, whenever he had a mathematical thought, he immediately wanted to write it down and elaborate.

ROTA : Did he have any students?

ULAM : Not really, even though at the Institute he gave several courses every year. Murray and Halperin may be considered his students.

ROTA : What about Gödel and von Neumann?

ULAM : One summer before the war when I was returning to the States, Johnny was waiting for me at the pier. His first words were that Gödel had shown that the continuum hypothesis was undecidable. This was how I heard for the first time about the existence of undecidable propositions in any formal system. So I said to him, "Oh! That is because he defines what is meant by a set." Johnny opened his eyes wide and expressed surprise that I had seen right away what was indeed the essential point. He thought I had some supernatural intuition.

I asked him whether Gödel was not a little afraid that his result was nothing but a sort of super paradox of the existing set theory, merely a diagonal method. In a sense it is a diagonalization. He agreed that this was probably right and that Gödel did not quite realise the importance of his discovery because of the fear that it would turn out to be merely another version of the whole series of set-theoretical paradoxes. Of course it was much more than that because he had made it all formal. The other paradoxes were special and dependent on metamathematical considerations that were not truly part of mathematics, whereas his results were. Curious how nervous people can be about their own work when it is the work!

ROTA : You have a higher opinion of Gödel than I have.

ULAM : Yes, I know. It was so unexpected at the time, and poor Hilbert was ...

ROTA : Not to speak of poor von Neumann.

ULAM : Johnny told me that Gödel's results made him very downcast, not quite despairing but disappointed. You must remember that his work on the axiomatisation of set theory, which was way back in the twenties, constitutes to this day one of the best foundations for set-theoretical mathematics. Basically he believed in Hilbert's goal of a final and conclusive axiomatisation of mathematics, and yet, in a 1925 paper, in a mysterious flash of intuition, he pointed out the limits of any axiomatic formulation of set theory. That was perhaps a vague forecast of Gödel's result. But it was left to Gödel to follow it through, and it has changed the direction of all science.

Gainesville
January 1974

Last Updated June 2024

DJF/JOC/EFR
Copyright information
Accessibility statement
School of Mathematics and Statistics
University of St Andrews, Scotland

University of St Andrews

If you have comments, or spot errors, we are always pleased to hear from you.